

Untitled

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2026-09-01

Data Understanding

Part 1: Just Pharmac Data

```
pharmac <- read.csv("Data_ByTG2_Regional.csv", header = TRUE)
head(pharmac)
```

```
##              Type TG2ID              TherapeuticGrp2
## 1 Initial dispensings    707 Agents Affecting the Renin-Angiotensin System
## 2 Initial dispensings    707 Agents Affecting the Renin-Angiotensin System
## 3 Initial dispensings    707 Agents Affecting the Renin-Angiotensin System
## 4 Initial dispensings    707 Agents Affecting the Renin-Angiotensin System
## 5 Initial dispensings    707 Agents Affecting the Renin-Angiotensin System
## 6 Initial dispensings    707 Agents Affecting the Renin-Angiotensin System
##      District YearDisp NumDisps NumPpl NHIComp
## 1      Auckland    2021   208874   55118    100
## 2   Bay of Plenty    2021   135111   34926    100
## 3     Canterbury    2021   266828   69100    100
## 4 Capital & Coast    2021   133210   34282    100
## 5 Counties Manukau    2021   299183   78608    100
## 6     Hawke's Bay    2021    95664   24700    100
```

```
# We can remove variables Type, TG2ID, and NHI Completeness
# Type and NHI Completeness only contain one unique variable,
# and TG2ID is a less informative repeat of TherapeuticGroup2
```

```
drops <- c("Type", "TG2ID", "NHIComp")
pharmac <- pharmac[ , !(names(pharmac) %in% drops)]

pharmac$TherapeuticGrp2 <- as.factor(pharmac$TherapeuticGrp2)
pharmac$District <- as.factor(pharmac$District)

head(pharmac)
```

```
##
## TherapeuticGrp2 District YearDisp
## 1 Agents Affecting the Renin-Angiotensin System Auckland 2021
## 2 Agents Affecting the Renin-Angiotensin System Bay of Plenty 2021
## 3 Agents Affecting the Renin-Angiotensin System Canterbury 2021
## 4 Agents Affecting the Renin-Angiotensin System Capital & Coast 2021
## 5 Agents Affecting the Renin-Angiotensin System Counties Manukau 2021
## 6 Agents Affecting the Renin-Angiotensin System Hawke's Bay 2021
## NumDisps NumPpl
## 1 208874 55118
## 2 135111 34926
## 3 266828 69100
## 4 133210 34282
## 5 299183 78608
## 6 95664 24700
```

```
dim(pharmac)
```

```
## [1] 12396 5
```

```
# 12396 observations with 5 variables
```

```
summary(pharmac)
```

```
##
## TherapeuticGrp2 District
## Agents Affecting the Renin-Angiotensin System: 105 New Zealand : 614
## Agents for Parkinsonism and Related Disorders: 105 Canterbury : 601
## Alpha-Adrenoceptor Blockers : 105 Waitemata : 601
## Anaesthetics : 105 Auckland : 599
## Analgesics : 105 Counties Manukau: 598
## Antacids and Antiflatulents : 105 Waikato : 598
## (Other) :11766 (Other) :8785
## YearDisp NumDisps NumPpl
## Min. :2021 Length:12396 Length:12396
## 1st Qu.:2022 Class :character Class :character
## Median :2023 Mode :character Mode :character
## Mean :2023
## 3rd Qu.:2024
## Max. :2025
##
```

```
# We can see NumDisp and NumPpl are characters when we want them to be numeric ints
# This is because observations with counts less than 6 are noted as <6
# to protect the anonymity of patients receiving less commonly prescribed medications
# We will have to deal with this.
```

```
# No missing data!
any(is.na(pharmac))
```

```
## [1] FALSE
```

```
# Shape and Structure of <6 observations
pharmacSuppressed <- pharmac %>% filter(NumDisps == "<6" | NumPpl == "<6")
head(pharmacSuppressed)
```

```
##              TherapeuticGrp2      District YearDisp NumDisps
## 1 Agents Used in the Treatment of Poisonings      Auckland      2021      <6
## 2 Agents Used in the Treatment of Poisonings      Bay of Plenty      2021        9
## 3 Agents Used in the Treatment of Poisonings      Capital & Coast      2021      <6
## 4 Agents Used in the Treatment of Poisonings      Counties Manukau      2021      <6
## 5 Agents Used in the Treatment of Poisonings      MidCentral      2021        7
## 6 Agents Used in the Treatment of Poisonings      Northland      2021      <6
##   NumPpl
## 1      <6
## 2      <6
## 3      <6
## 4      <6
## 5      <6
## 6      <6
```

```
dim(pharmacSuppressed)
```

```
## [1] 524    5
```

```
# 524 suppressed observations
```

```
summary(pharmacSuppressed)
```

```
##              TherapeuticGrp2      District
## Respiratory Stimulants      : 54      South Canterbury: 42
## Agents Used in the Treatment of Poisonings: 48      Wairarapa      : 40
## Food Thickeners      : 48      West Coast      : 40
## Gluten Free Foods      : 46      Tairawhiti      : 34
## Ketogenic Diet      : 45      Whanganui      : 34
## Immune Modulators      : 40      Taranaki      : 33
## (Other)      :243      (Other)      :301
##   YearDisp      NumDisps      NumPpl
## Min.      :2021      Length:524      Length:524
## 1st Qu.:2021      Class :character      Class :character
## Median :2023      Mode  :character      Mode  :character
## Mean      :2023
## 3rd Qu.:2024
## Max.      :2025
##
```

```
# TG2 pharmaceuticals with most commonly suppressed observations
topSuppressed <- pharmacSuppressed %>%
  count(TherapeuticGrp2, sort = TRUE, name = "TG2_count") %>%
  slice_head(n = 10)

print(topSuppressed)
```

```
##              TherapeuticGrp2 TG2_count
## 1          Respiratory Stimulants      54
## 2    Agents Used in the Treatment of Poisonings      48
## 3              Food Thickeners      48
## 4          Gluten Free Foods      46
## 5          Ketogenic Diet      45
## 6          Immune Modulators      40
## 7 Foods And Supplements For Inherited Metabolic Disease      39
## 8          Detection of Substances in Urine      33
## 9          Antimalarials      23
## 10         Antifibrotics      21
```

```
# Districts with most commonly suppressed observations
topSuppressed <- pharmacSuppressed %>%
  count(District, sort = TRUE, name = "District_count") %>%
  slice_head(n = 10)

print(topSuppressed)
```

```
##              District District_count
## 1    South Canterbury      42
## 2          Wairarapa      40
## 3          West Coast      40
## 4          Tairāwhiti      34
## 5          Whanganui      34
## 6          Taranaki      33
## 7 Nelson Marlborough      31
## 8          Bay of Plenty      27
## 9              Lakes      27
## 10         Hawke's Bay      26
```

```
# Now that we've identified rows with <6 observations, we can remove these rows
# from the working dataset and convert NumPpl and NumDisps to int
```

```
pharmac <- pharmac %>%
  filter(NumDisps != "<6", NumPpl != "<6") %>%

  mutate(
    across(c(NumDisps, NumPpl), as.numeric)
  )
```

```
# Yahoo!!
summary(pharmac)
```

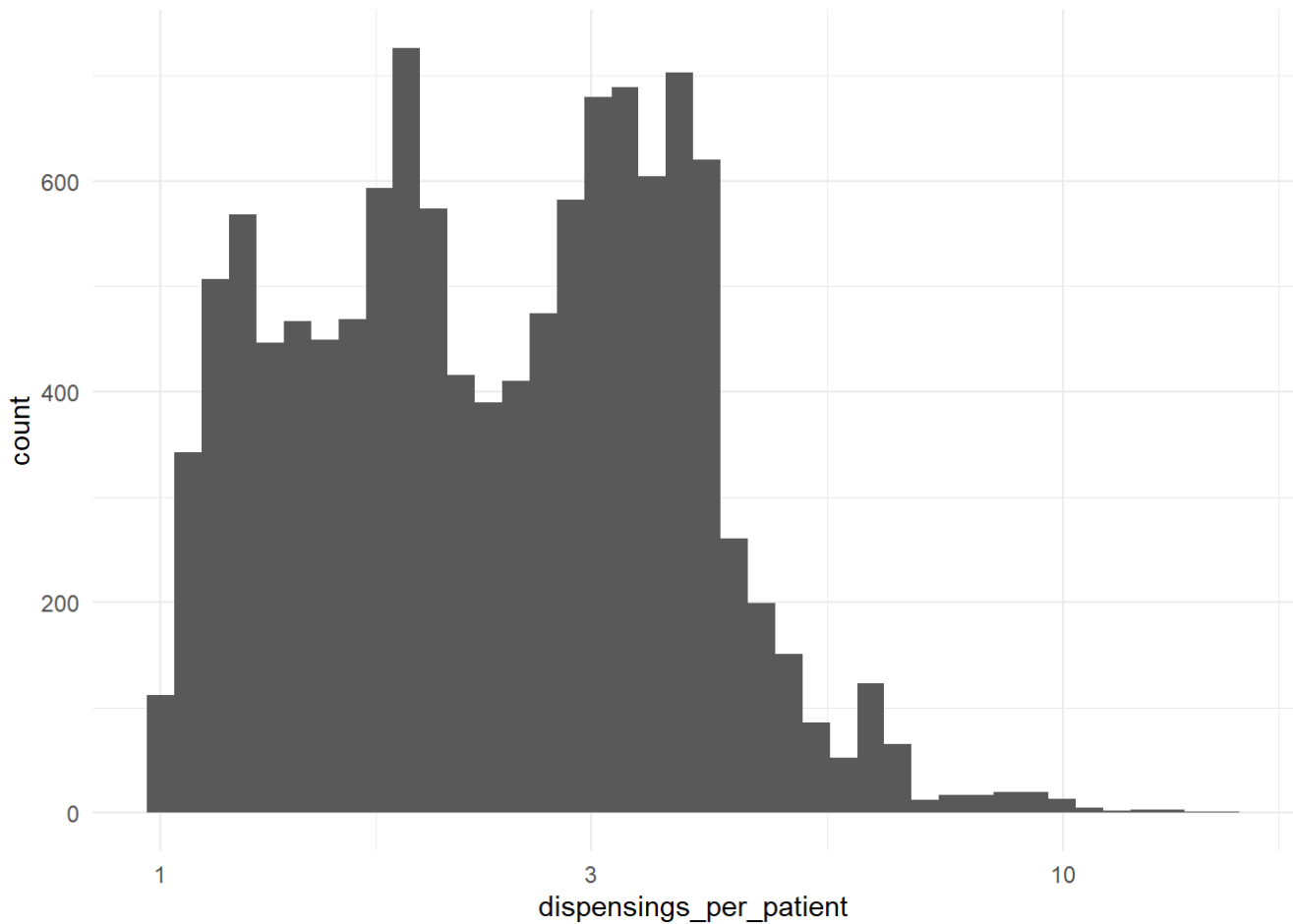
```
##
## TherapeuticGrp2 District
## Agents Affecting the Renin-Angiotensin System: 105 New Zealand : 600
## Agents for Parkinsonism and Related Disorders: 105 Waitemata : 590
## Alpha-Adrenoceptor Blockers : 105 Canterbury : 588
## Anaesthetics : 105 Counties Manukau: 584
## Analgesics : 105 Waikato : 584
## Antacids and Antiflatulents : 105 Auckland : 580
## (Other) :11242 (Other) :8346
## YearDisp NumDisps NumPpl
## Min. :2021 Min. : 6 Min. : 6
## 1st Qu.:2022 1st Qu.: 889 1st Qu.: 390
## Median :2023 Median : 5496 Median : 2530
## Mean :2023 Mean : 46277 Mean : 17980
## 3rd Qu.:2024 3rd Qu.: 24532 3rd Qu.: 10146
## Max. :2025 Max. :5833774 Max. :2059390
##
```

```
pharmacCorr <- pharmac %>%
  mutate(
    dispensings_per_patient = NumDisps / NumPpl
  )
```

```
summary(pharmacCorr$dispensings_per_patient)
```

```
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1.000 1.594 2.333 2.616 3.388 15.143
```

```
ggplot(pharmacCorr, aes(x = dispensings_per_patient)) +
  geom_histogram(bins = 40) +
  scale_x_log10() +
  theme_minimal()
```



```
pharmacCorr %>%
  group_by(TherapeuticGrp2) %>%
  summarise(
    median_dispensings_per_patient =
      median(dispensings_per_patient, na.rm = TRUE)
  ) %>%
  slice_max(median_dispensings_per_patient, n = 15)
```

```
## # A tibble: 15 × 2
##   TherapeuticGrp2                median_dispensings_pe...1
##   <fct>                                <dbl>
## 1 Foods And Supplements For Inherited Metabolic Disease      8.69
## 2 Stimulants/ADHD Treatments                                7.46
## 3 Diabetes                                                      6.21
## 4 Agents for Parkinsonism and Related Disorders              5.90
## 5 Antiretrovirals                                              5.34
## 6 Thyroid and Antithyroid Agents                             4.76
## 7 Antipsychotics                                              4.68
## 8 Immunosuppressants                                          4.42
## 9 Antihypotensives                                           4.24
## 10 Treatments for Dementia                                    4.23
## 11 Diabetes Management                                        4.14
## 12 Diuretics                                                  4.10
## 13 Antiepilepsy Drugs                                         3.98
## 14 Chemotherapeutic Agents                                    3.97
## 15 Beta-Adrenoceptor Blockers                                 3.97
## # i abbreviated name: 1median_dispensings_per_patient
```

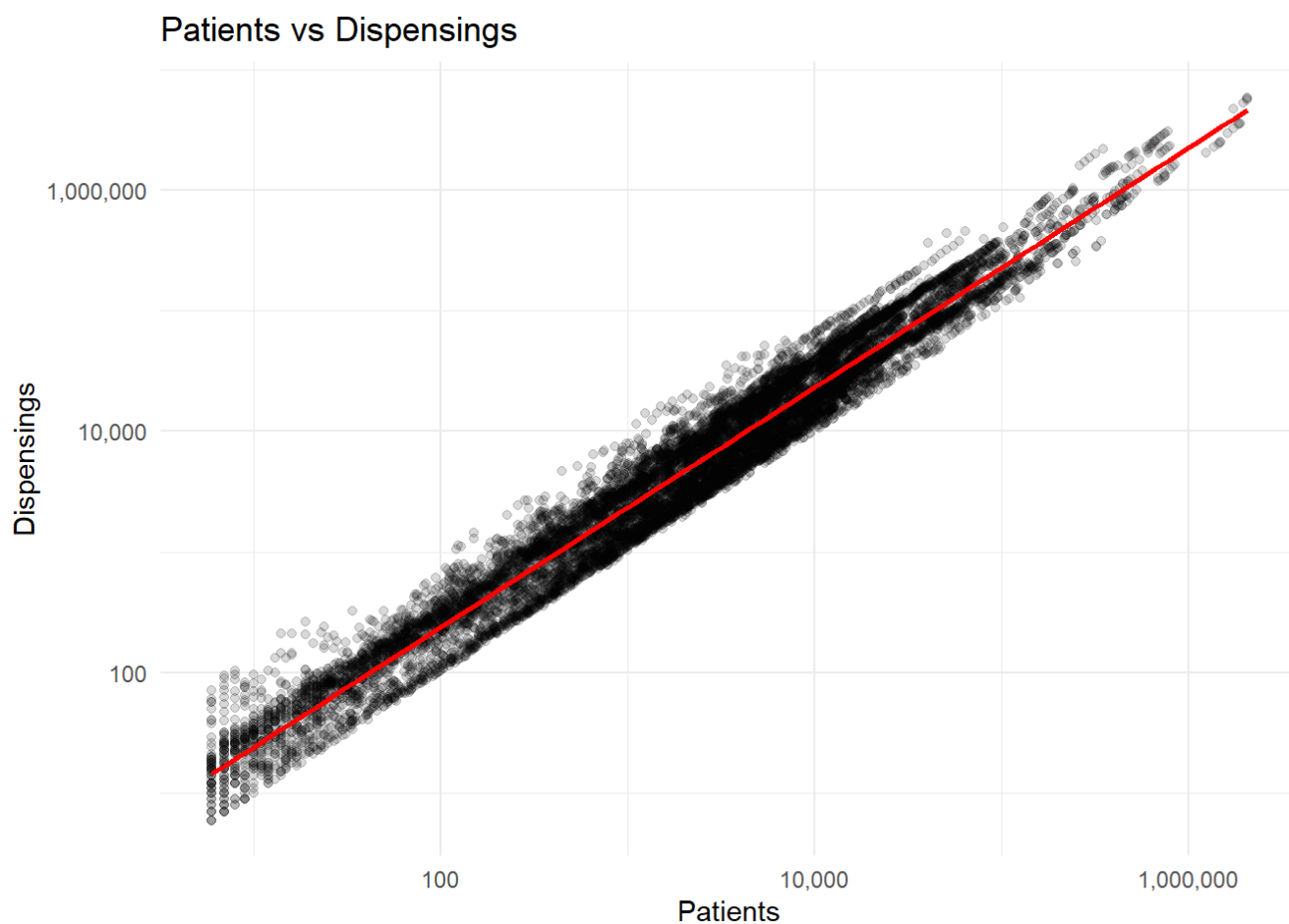
```
pharmacCorr %>%  
  summarise(  
    correlation = cor(NumPpl, NumDisps, use = "complete.obs")  
  )
```

```
## correlation  
## 1 0.9473452
```

```
# Surprise, highly correlated
```

```
ggplot(pharmacCorr, aes(x = NumPpl, y = NumDisps)) +  
  geom_point(alpha = 0.15) +  
  scale_x_log10(labels = scales::comma) +  
  scale_y_log10(labels = scales::comma) +  
  geom_smooth(method = "lm", se = FALSE, colour = "red") +  
  labs(  
    title = "Patients vs Dispensings",  
    x = "Patients",  
    y = "Dispensings"  
  ) +  
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



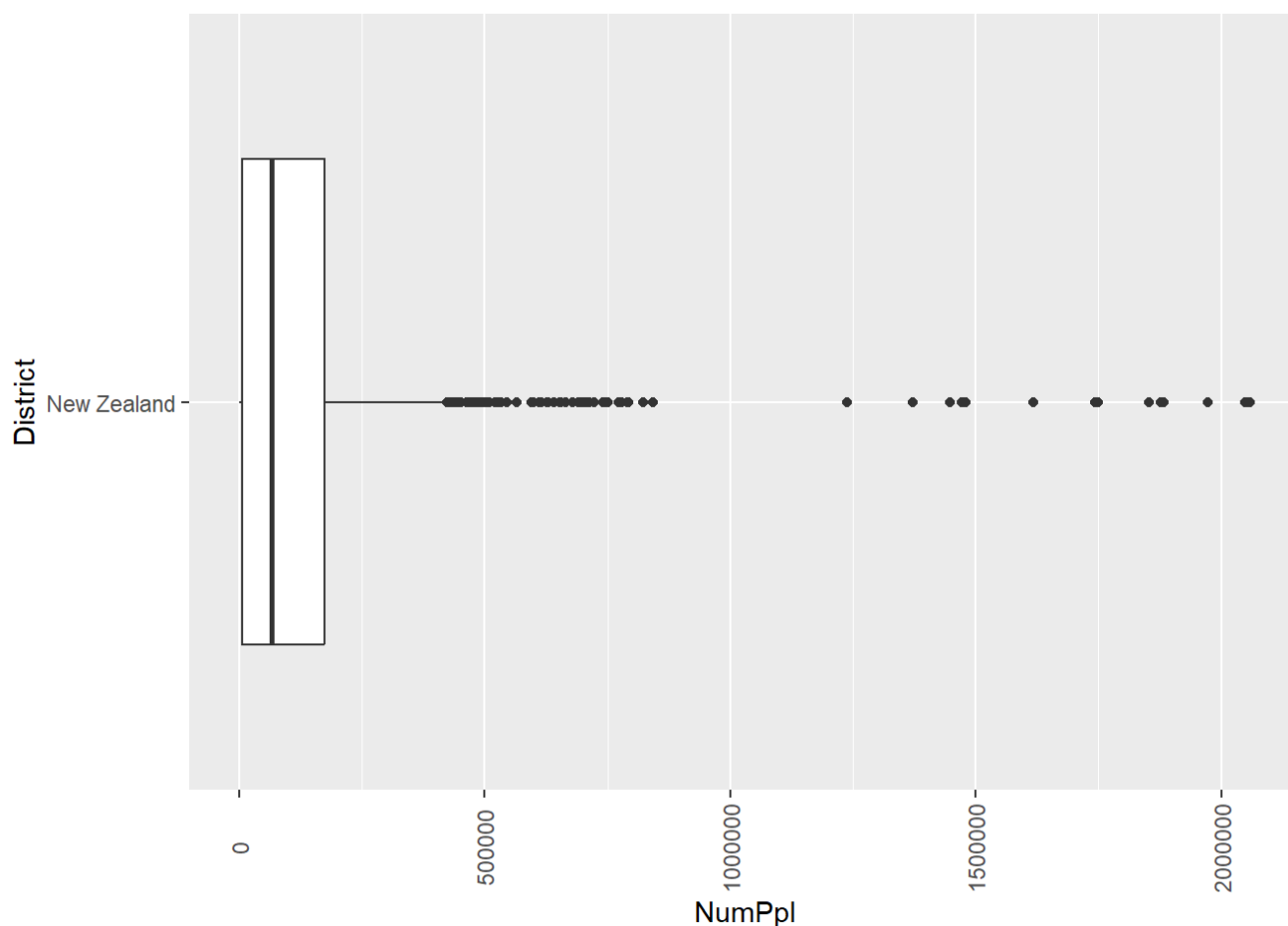
```
# NumPpl and NumDisps are expected to be strongly positively associated because more patients generally generate more dispensings
```

```
# Now we actually need to do EDA...
```

```
# First correlation plot. Only 2 numeric variables and they should correlate with each other very strongly due to multicollinearity
```

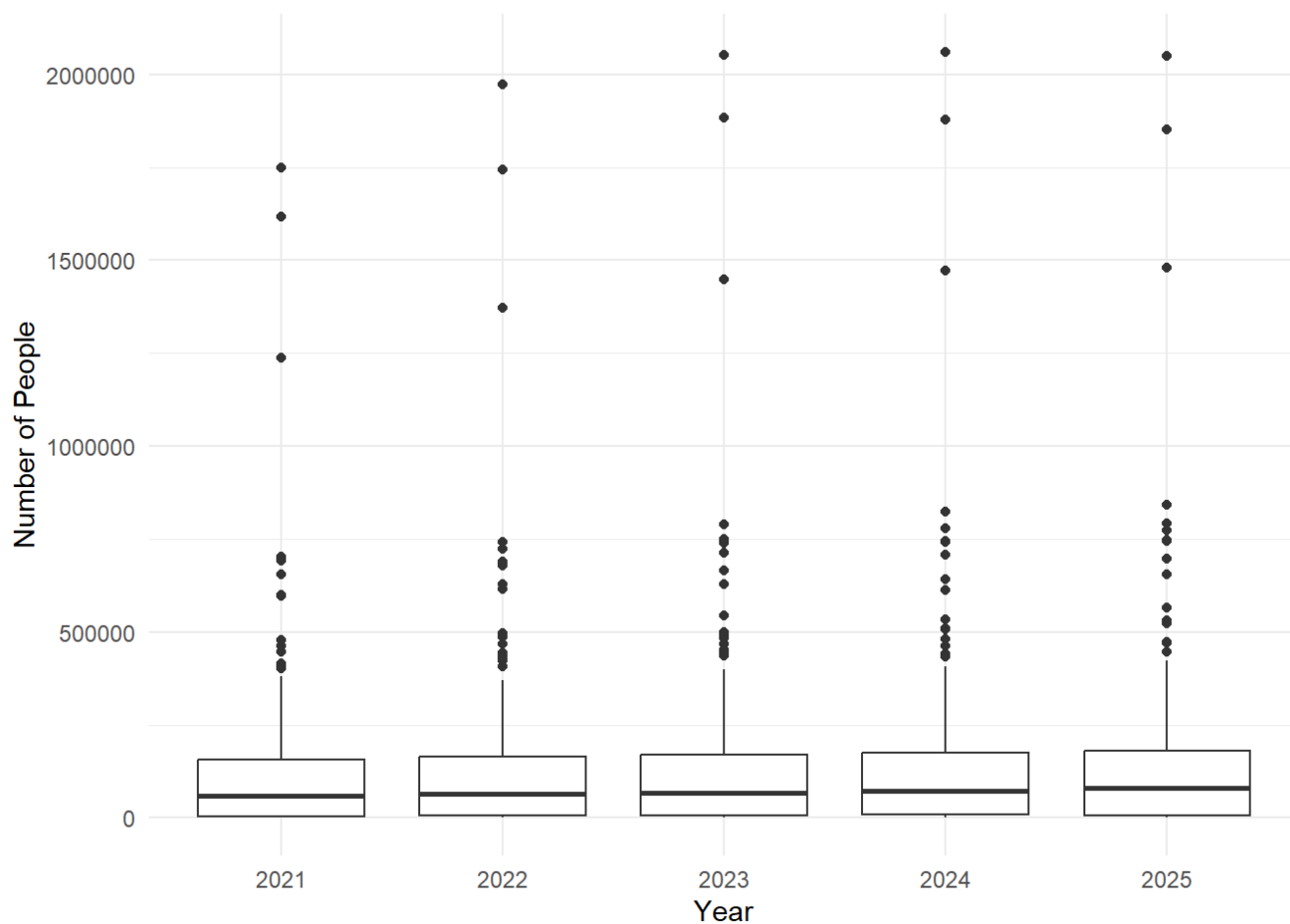
```
pharmacNZ <- pharmac %>% filter(District == "New Zealand")
```

```
ggplot(pharmacNZ, aes(x=NumPpl, y=District)) + geom_boxplot() + theme(axis.text.x = element_text(angle=90, vjust=0.5))
```



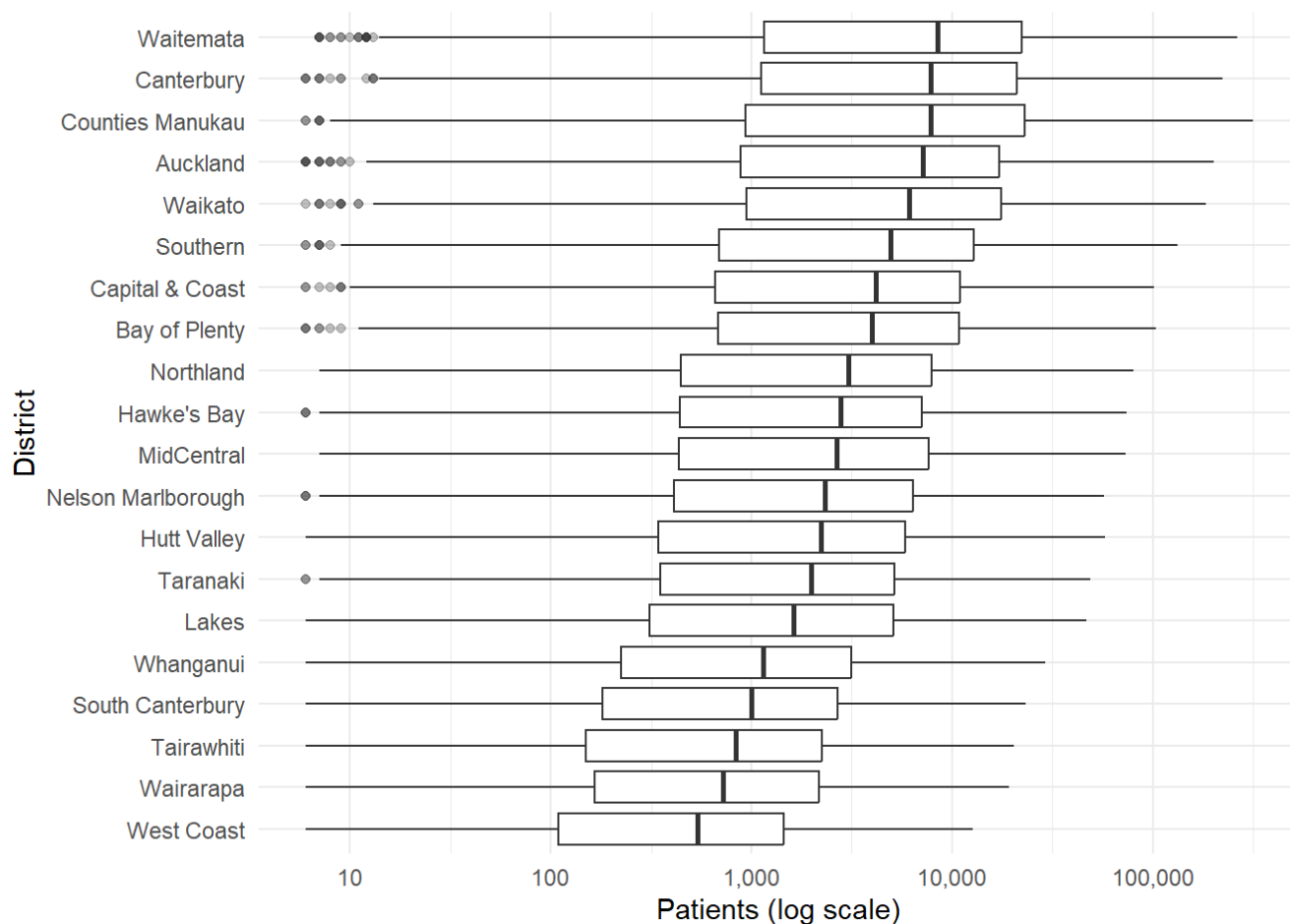
```
# New Zealand totals per year
```

```
ggplot(pharmacNZ, aes(x = factor(YearDisp), y = NumPpl)) +  
  geom_boxplot() +  
  labs(  
    x = "Year",  
    y = "Number of People"  
  ) +  
  theme_minimal()
```

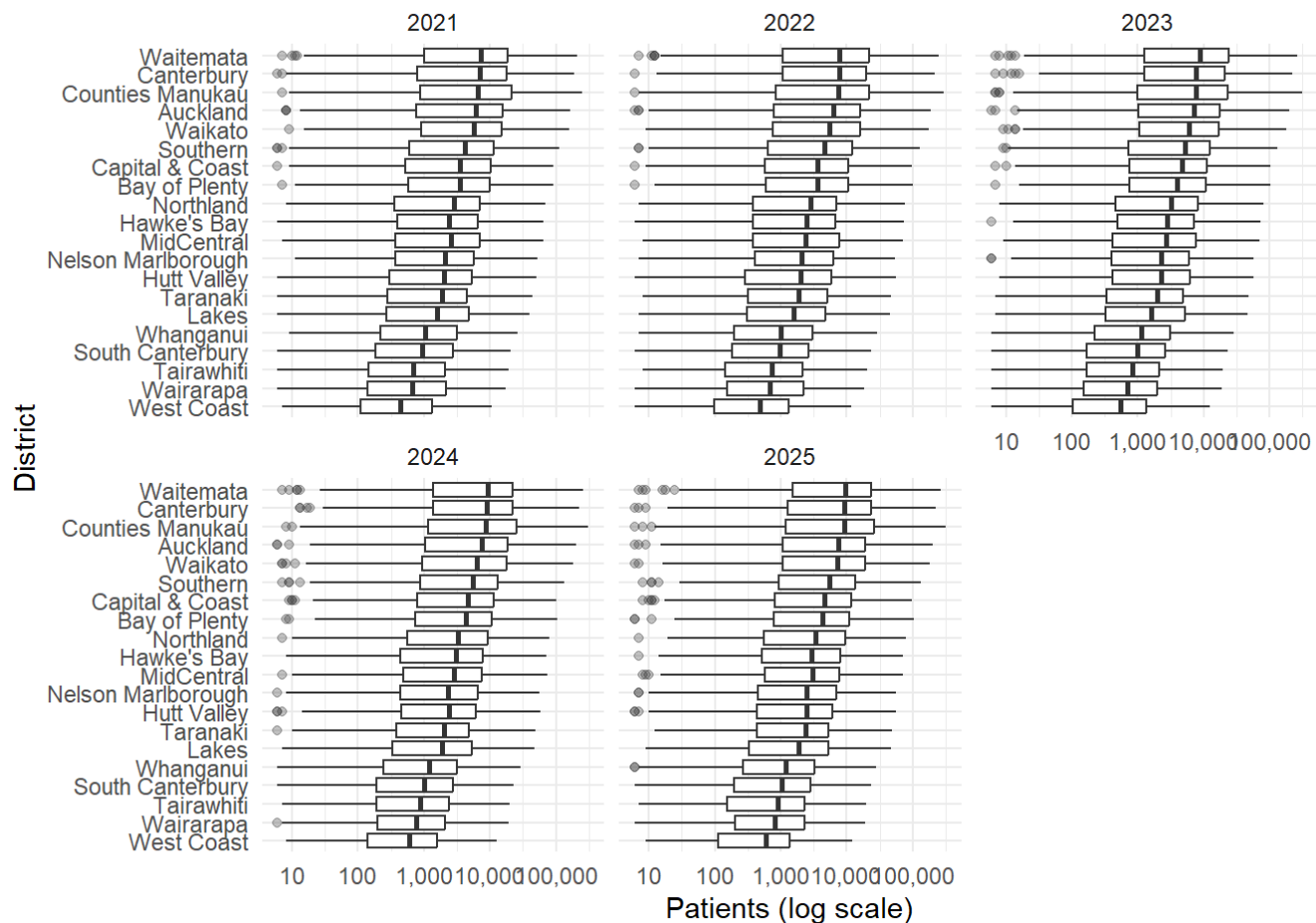



```
pharmacDistrict <- pharmac %>% filter(District != "New Zealand")

ggplot(pharmacDistrict, aes(x = reorder(District, NumPpl, median), y = NumPpl)) +
  geom_boxplot(outlier.alpha = 0.3) +
  coord_flip() +
  scale_y_log10(labels = scales::comma) +
  labs(
    x = "District",
    y = "Patients (log scale)"
  ) +
  theme_minimal()
```



```
ggplot(pharmacDistrict, aes(x = reorder(District, NumPpl, median), y = NumPpl)) +
  geom_boxplot(outlier.alpha = 0.3) +
  facet_wrap(~ YearDisp) +
  coord_flip() +
  scale_y_log10(labels = scales::comma) +
  labs(
    x = "District",
    y = "Patients (log scale)"
  ) +
  theme_minimal()
```

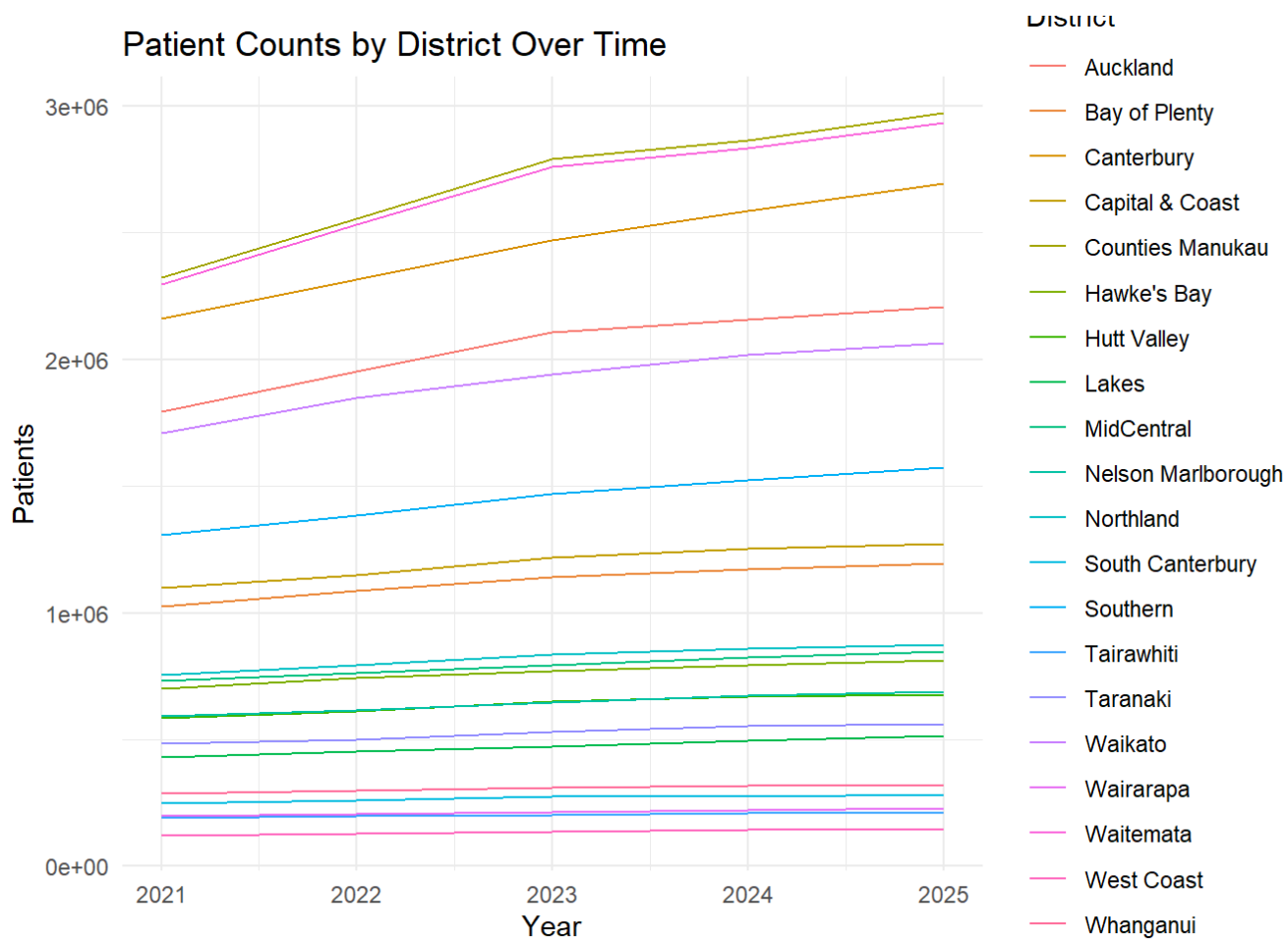


```

pharmacDistrict %>%
  group_by(District, YearDisp) %>%
  summarise(
    patients = sum(NumPpl),
    dispensings = sum(NumDisps),
    .groups = "drop"
  ) %>%
  ggplot(aes(x = YearDisp, y = patients, colour = District)) +
  geom_line() +
  labs(
    title = "Patient Counts by District Over Time",
    x = "Year",
    y = "Patients"
  ) +
  theme_minimal()

```

Patient Counts by District Over Time



```

top10_grps <- pharmacDistrict %>%
  count(TherapeuticGrp2, wt = NumDisps, name = "total_dispensings") %>%
  slice_max(total_dispensings, n = 10, with_ties = FALSE) %>%
  pull(TherapeuticGrp2)

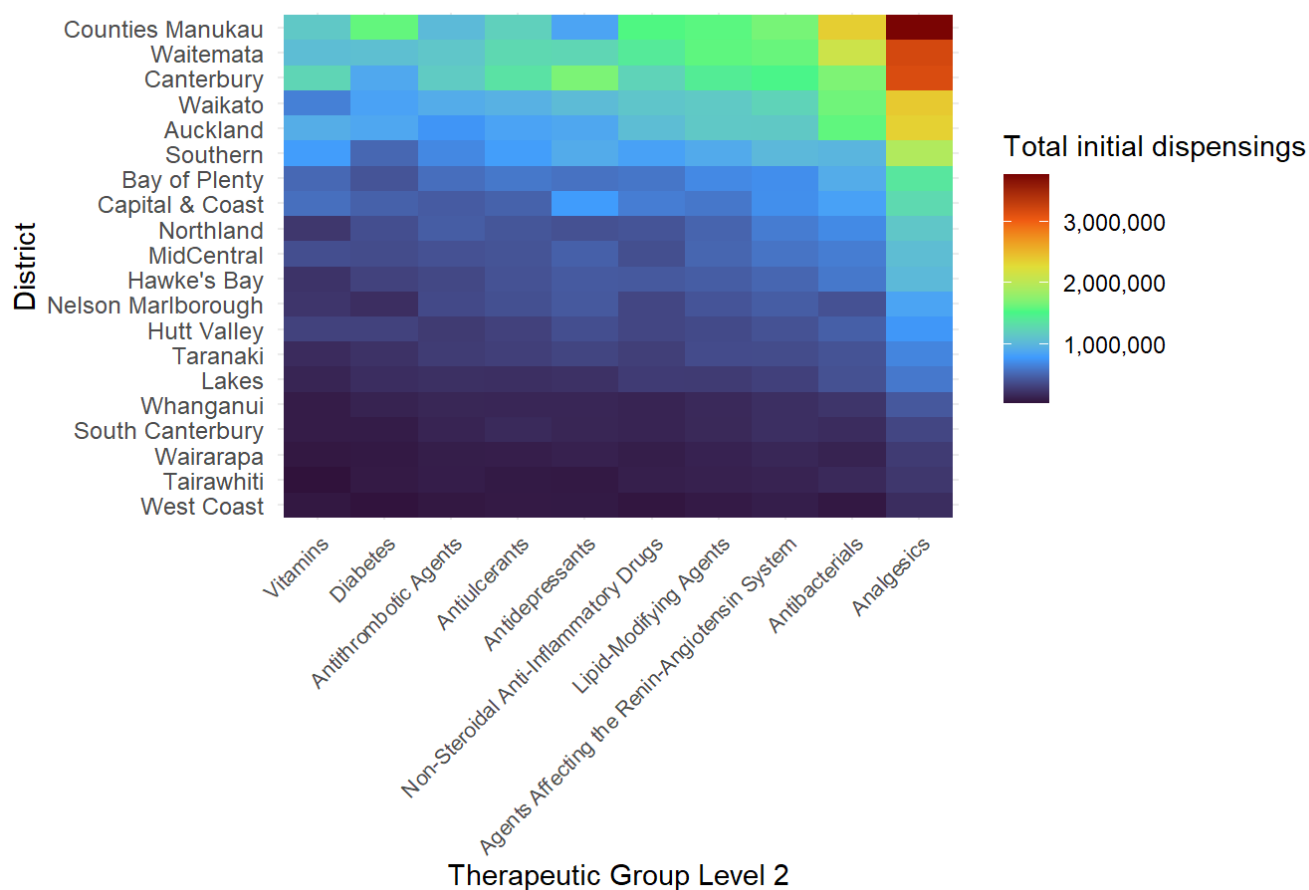
pharmacDistrict %>%
  filter(TherapeuticGrp2 %in% top10_grps) %>%
  group_by(District, TherapeuticGrp2) %>%
  summarise(total_dispensings = sum(NumDisps, na.rm = TRUE), .groups = "drop") %>%

  mutate(
    TherapeuticGrp2 = factor(TherapeuticGrp2, levels = rev(top10_grps)),
    District = fct_reorder(District, total_dispensings, .fun = sum)
  ) %>%

  ggplot(aes(x = TherapeuticGrp2, y = District, fill = total_dispensings)) +
  geom_tile() +
  scale_fill_viridis_c(option = "turbo", labels = scales::comma) +
  labs(
    title = "Top 10 Therapeutic Groups by District",
    x = "Therapeutic Group Level 2",
    y = "District",
    fill = "Total initial dispensings"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(size = 14, face = "bold"),
    axis.text.x = element_text(angle = 45, hjust = 1, size = 8),
    axis.text.y = element_text(size = 9)
  )

```

Top 10 Therapeutic Groups by District



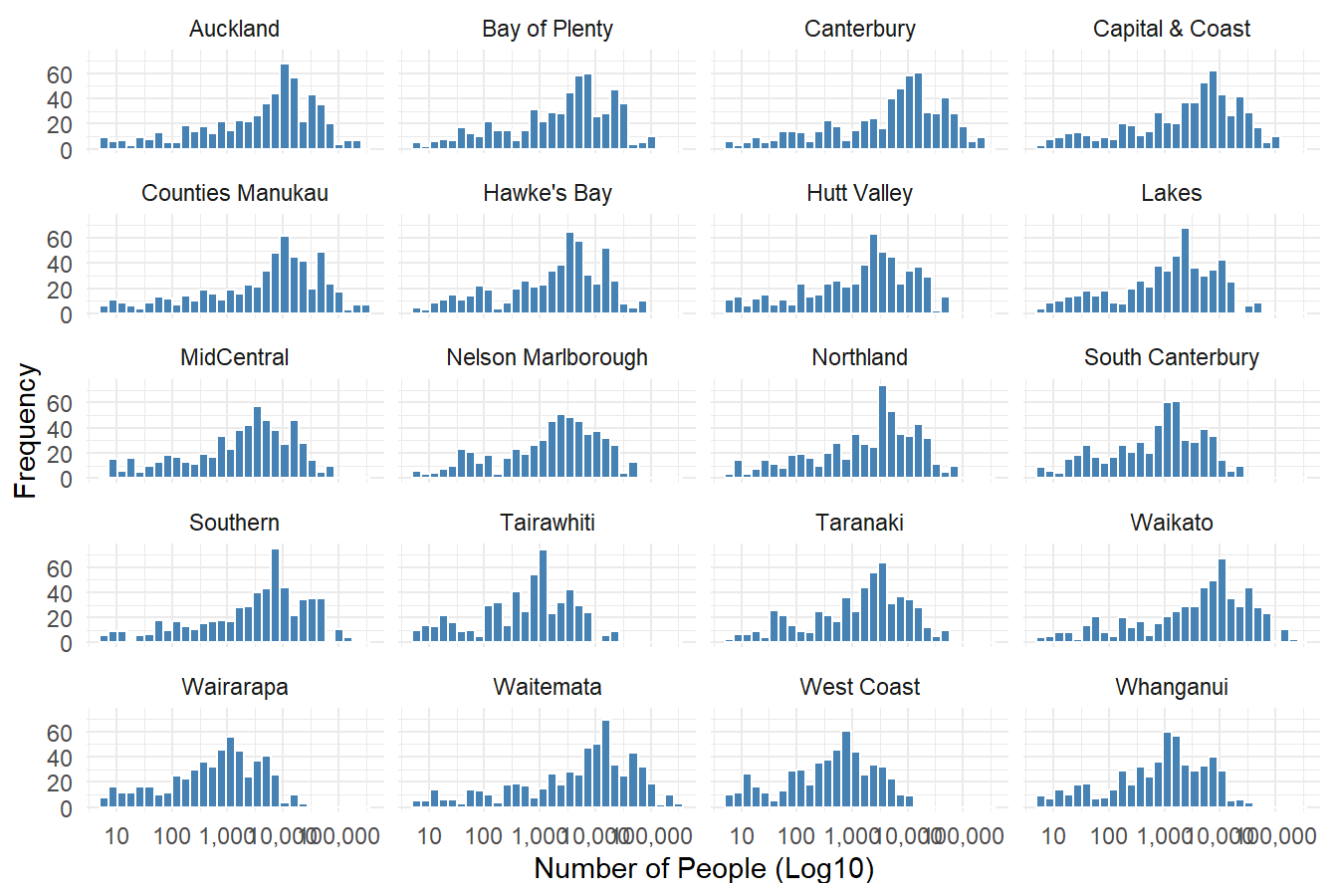
Number of bins for a histogram

```
bins <- as.integer(sqrt(nrow(pharmac)))
bins
```

```
## [1] 108
```

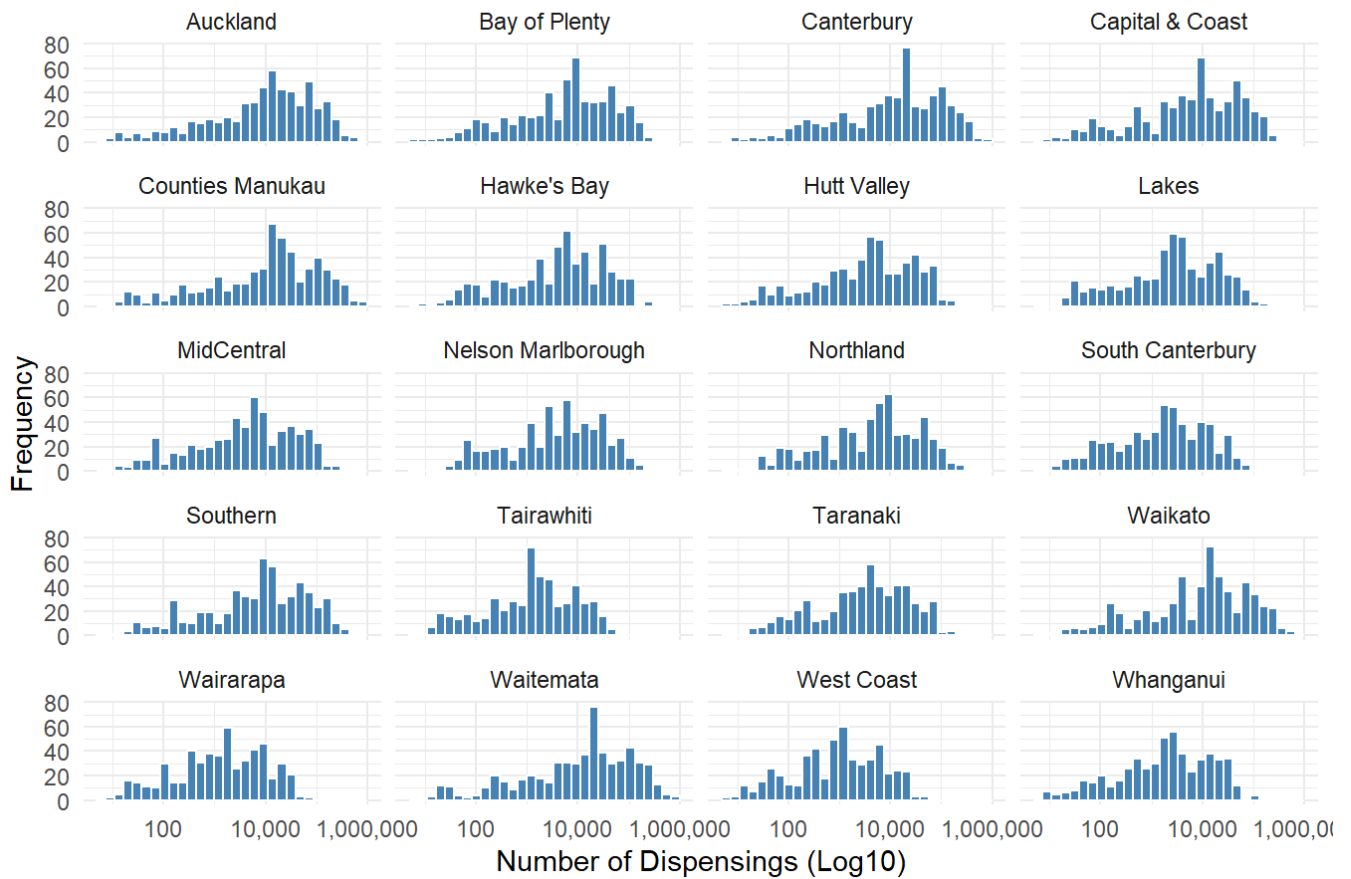
```
ggplot(pharmacDistrict, aes(x = NumPpl)) +
  geom_histogram(bins = 30, fill = "steelblue", color = "white") +
  scale_x_log10(labels = scales::comma) + # Essential for heavily skewed count data
  labs(title = "Distribution of Patients per District (Log Scale)", x = "Number of People (Log10)", y = "Frequency") +
  facet_wrap(~ District, ncol = 4) +
  theme_minimal()
```

Distribution of Patients per District (Log Scale)



```
ggplot(pharmacDistrict, aes(x = NumDisps)) +
  geom_histogram(bins = 30, fill = "steelblue", color = "white") +
  scale_x_log10(labels = scales::comma) + # Essential for heavily skewed count data
  labs(title = "Distribution of Dispensings per District (Log Scale)", x = "Number of Dispensings (Log10)", y = "Frequency") +
  facet_wrap(~ District, ncol = 4) +
  theme_minimal()
```

Distribution of Dispensings per District (Log Scale)



Part 2: Just Census Data

```
census <- read.csv("census2023.csv", header = TRUE)
head(census)
```



```
##           Area Population Age_Under15 Age_15to29 Age_30to64 Age_Over65
## 1      New Zealand   5090715      936300      968304      2260728      828588
## 2      Northland    196464      38067      29856      83790      42288
## 3      Waitemata    637884      121626      117051      300351      92355
## 4      Auckland    480438      73830      109974      219471      61671
## 5 Counties Manukau 591720      128781      124209      263157      69045
## 6      Waikato     443433      88611      83058      190581      75789
## Ethnicity_European Ethnicity_Maori Ethnicity_Pasifika Ethnicity_Asian
## 1      3383742      887493      442632      861576
## 2      141591      72612      9486      9345
## 3      387063      69090      58188      176328
## 4      239460      43455      61647      156603
## 5      216600      96411      157086      188715
## 6      312012      110580      22992      54807
## Ethnicity_MELAAF Ethnicity_Other Maori_Descent No_Maori_Descent Gender_Male
## 1      92760      56133      978246      3873726      2460741
## 2      1263      2208      77472      110064      95697
## 3      17334      6729      79266      537720      310971
## 4      17229      3645      49857      406002      229092
## 5      10494      5169      104874      467121      290775
## 6      6528      4893      121026      303255      216240
## Gender_Female Gender_Other Children_Average Smoking_Regular Smoking_ExSmoker
## 1      2512779      20403      1.6      310809      1012422
## 2      97776      528      1.9      16644      51108
## 3      318192      2226      1.5      28317      111828
## 4      233430      2424      1.2      21606      72024
## 5      292968      1449      1.7      40131      86403
## 6      220278      1524      1.8      31296      92154
## Smoking_NonSmoker Not_Disabled Disabled Income Employment_FullTime
## 1      2734392      3695919      298980      41500      2077071
## 2      88182      124536      14049      33100      67788
## 3      369609      487782      30501      46300      272370
## 4      297486      361407      22542      47700      211368
## 5      329874      408966      31152      41300      239712
## 6      225975      312288      28269      39700      174627
## Employment_PartTime Employment_Unemployed Employment_NotInLabourForce
## 1      545649      120129      1314774
## 2      21735      5469      60945
## 3      66531      14112      156747
## 4      49482      12708      117561
## 5      48576      17589      150531
## 6      46758      11163      116880
## HoursWorked
## 1      37.2
## 2      36.5
## 3      37.1
## 4      37.4
## 5      38.2
## 6      37.7
```

```
dim(census)
```

```
## [1] 20 29
```

```
# 20 values with 29 variables.
```

```
# New Zealand is included, Capital and Coast and Hutt Valley are combined into one area (r  
epresents merger of the two districts in 2021)
```

```
summary(census)
```

```

##      Area                Population      Age_Under15      Age_15to29
## Length:20              Min.    : 36408    Min.    : 5448    Min.    : 4518
## Class :character       1st Qu.: 106159    1st Qu.: 20996    1st Qu.: 18172
## Mode  :character       Median : 191223    Median : 36791    Median : 32588
##                               Mean  : 509010    Mean  : 93629    Mean  : 96829
##                               3rd Qu.: 469114    3rd Qu.: 81368    3rd Qu.:102621
##                               Max.   :5090715    Max.   :936300    Max.   :968304
##      Age_30to64      Age_Over65      Ethnicity_European Ethnicity_Maori
## Min.    : 15861    Min.    : 7560    Min.    : 28905    Min.    : 4491
## 1st Qu.: 45089    1st Qu.: 18506    1st Qu.: 70097    1st Qu.: 25617
## Median : 81452    Median : 39479    Median : 143487    Median : 44052
## Mean   : 226068    Mean   : 82855    Mean   : 338365    Mean   : 88748
## 3rd Qu.: 216238    3rd Qu.: 67567    3rd Qu.: 296325    3rd Qu.: 69617
## Max.   :2260728    Max.   :828588    Max.   :3383742    Max.   :887493
## Ethnicity_Pasifika Ethnicity_Asian Ethnicity_MELAAF Ethnicity_Other
## Min.    : 540      Min.    : 1335    Min.    : 180.0    Min.    : 468
## 1st Qu.: 3182      1st Qu.: 6356    1st Qu.: 765.8    1st Qu.: 1190
## Median : 9651      Median : 13869    Median : 1966.5    Median : 2391
## Mean   : 44262      Mean   : 86158    Mean   : 9276.3    Mean   : 5614
## 3rd Qu.: 28503      3rd Qu.: 78206    3rd Qu.: 9866.2    3rd Qu.: 4927
## Max.   :442632      Max.   :861576    Max.   :92760.0    Max.   :56133
## Maori_Descent      No_Maori_Descent Gender_Male      Gender_Female
## Min.    : 5085      Min.    : 21120    Min.    : 17025    Min.    : 16245
## 1st Qu.: 28648      1st Qu.: 60748    1st Qu.: 50744    1st Qu.: 51714
## Median : 51660      Median : 131952    Median : 93111    Median : 95648
## Mean   :111553      Mean   : 387364    Mean   : 246067    Mean   : 251275
## 3rd Qu.: 77986      3rd Qu.: 383243    3rd Qu.: 225470    3rd Qu.: 233358
## Max.   :978246      Max.   :3873726    Max.   :2460741    Max.   :2512779
## Gender_Other      Children_Average Smoking_Regular Smoking_ExSmoker
## Min.    : 117.0      Min.    :1.20      Min.    : 3252    Min.    : 9066
## 1st Qu.: 303.0      1st Qu.:1.60      1st Qu.: 8350    1st Qu.: 24734
## Median : 664.5      Median :1.80      Median : 15594    Median : 46562
## Mean   : 2040.3      Mean   :1.73      Mean   : 31080    Mean   : 101238
## 3rd Qu.: 1699.5      3rd Qu.:1.90      3rd Qu.: 24690    3rd Qu.: 87841
## Max.   :20403.0      Max.   :2.00      Max.   :310809    Max.   :1012422
## Smoking_NonSmoker Not_Disabled      Disabled      Income
## Min.    : 15618      Min.    : 23664    Min.    : 2631    Min.    :32700
## 1st Qu.: 48684      1st Qu.: 67957    1st Qu.: 7160    1st Qu.:37875
## Median : 90893      Median : 129387    Median : 13918    Median :39200
## Mean   : 273434      Mean   : 369583    Mean   : 29897    Mean   :39625
## 3rd Qu.: 274050      3rd Qu.: 361858    3rd Qu.: 27824    3rd Qu.:41350
## Max.   :2734392      Max.   :3695919    Max.   :298980    Max.   :50400
## Employment_FullTime Employment_PartTime Employment_Unemployed
## Min.    : 12819      Min.    : 4101      Min.    : 687
## 1st Qu.: 40314      1st Qu.: 11219      1st Qu.: 2365
## Median : 71298      Median : 21354      Median : 4887
## Mean   : 207703      Mean   : 54564      Mean   : 12013
## 3rd Qu.: 211950      3rd Qu.: 48803      3rd Qu.: 11500
## Max.   :2077071      Max.   :545649      Max.   :120129
## Employment_NotInLabourForce HoursWorked
## Min.    : 10335      Min.    :36.40
## 1st Qu.: 27305      1st Qu.:36.75
## Median : 56376      Median :37.20
## Mean   : 131474      Mean   :37.19

```

```
## 3rd Qu.: 117050          3rd Qu.:37.40
## Max.    :1314774        Max.    :38.40
```

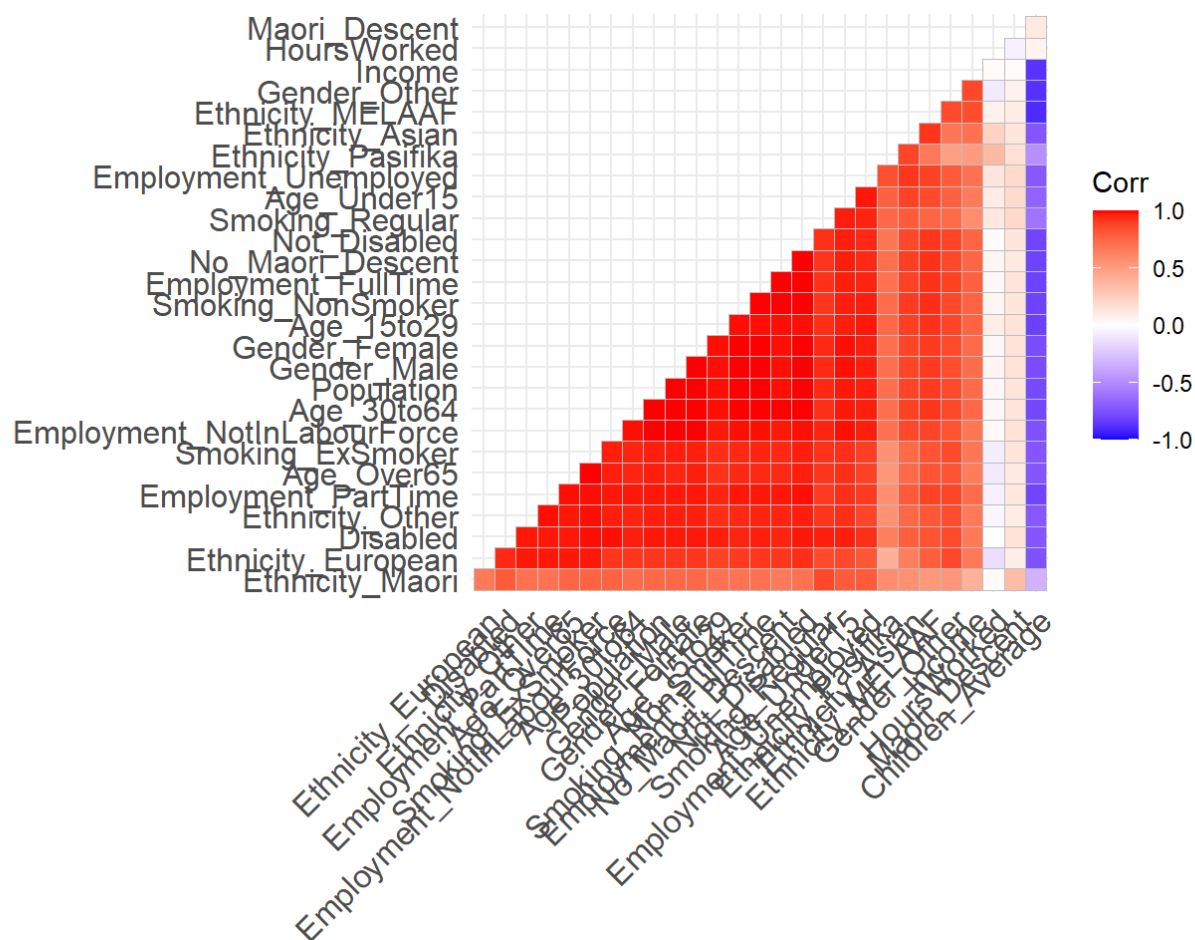
```
# No missing data!
any(is.na(census))
```

```
## [1] FALSE
```

```
census_numeric <- census %>%
  filter(Area != "New Zealand") %>%
  select(where(is.numeric))

cor_matrix <- cor(census_numeric, use = "pairwise.complete.obs")

ggcorrplot(
  cor_matrix,
  type = "lower",
  lab = FALSE,
  hc.order = TRUE
)
```



```

area_summaries <- census %>%
  filter(Area != "New Zealand") %>%
  select(Area, where(is.numeric)) %>%
  pivot_longer(cols = -Area, names_to = "Variable", values_to = "Value") %>%
  group_by(Area, Variable) %>%
  summarise(Mean_Value = mean(Value, na.rm = TRUE), .groups = "drop")

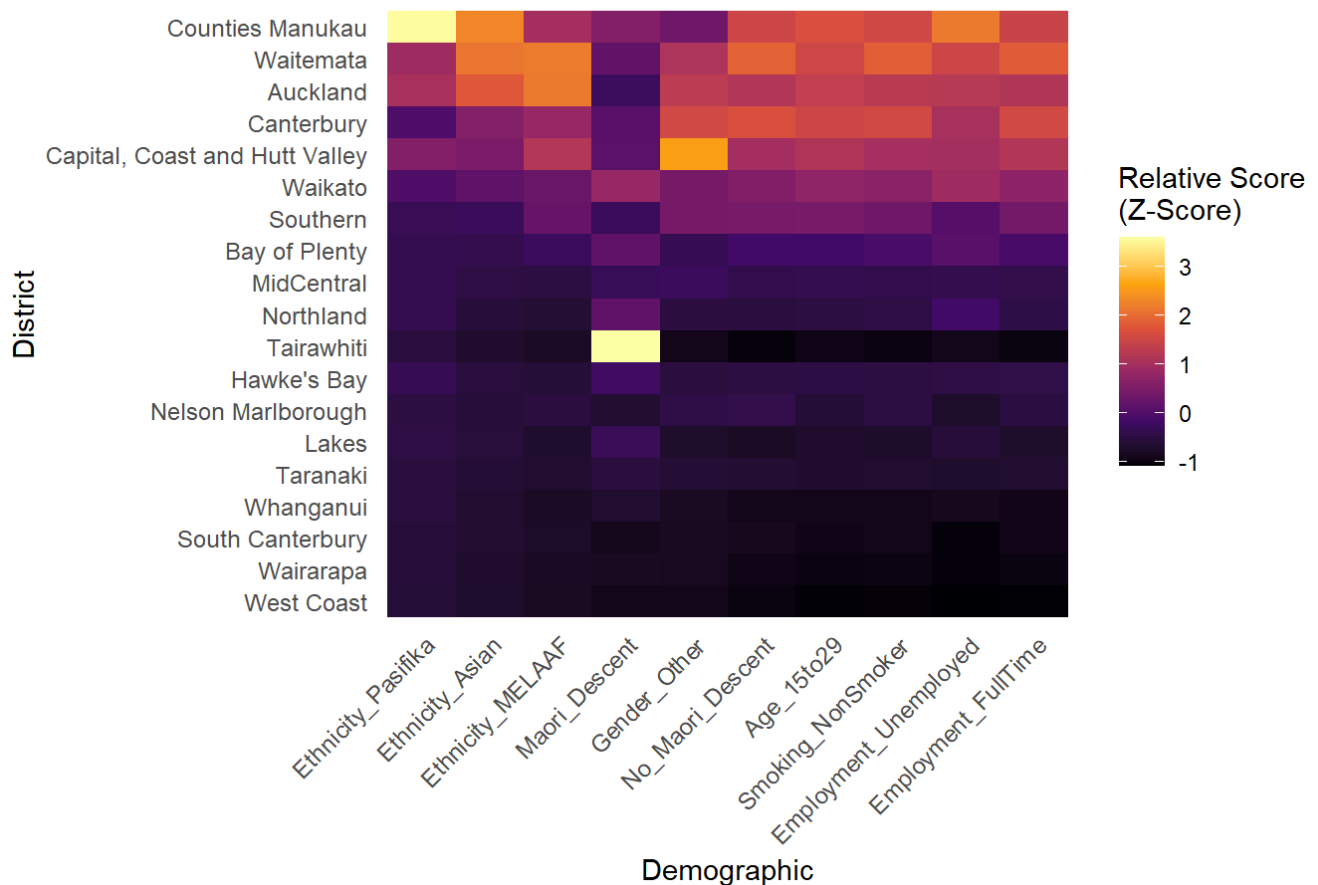
top10_differentiating_vars <- census %>%
  filter(Area != "New Zealand") %>%
  select(where(is.numeric)) %>%
  summarise(across(everything(), ~ sd(.x, na.rm = TRUE) / mean(.x, na.rm = TRUE))) %>%
  pivot_longer(cols = everything(), names_to = "Variable", values_to = "CV") %>%
  slice_max(CV, n = 10, with_ties = FALSE) %>%
  pull(Variable)

area_summaries %>%
  filter(Variable %in% top10_differentiating_vars) %>%
  group_by(Variable) %>%
  mutate(Scaled_Value = scale(Mean_Value)[, 1]) %>%
  ungroup() %>%
  mutate(
    Variable = factor(Variable, levels = top10_differentiating_vars),
    Area = fct_reorder(Area, Scaled_Value, .fun = sum)
  ) %>%

  ggplot(aes(x = Variable, y = Area, fill = Scaled_Value)) +
  geom_tile(linewidth = 0.1) +
  scale_fill_viridis_c(option = "inferno", name = "Relative Score\n(Z-Score)") +
  labs(
    title = "Top 10 Demographic Profiles in Districts",
    x = "Demographic",
    y = "District"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(size = 14, face = "bold"),
    axis.text.x = element_text(angle = 45, hjust = 1, size = 9),
    axis.text.y = element_text(size = 9),
    panel.grid = element_blank()
  )

```

Top 10 Demographic Profiles in Districts



Part 3: Putting Them Together

```
to_merge <- pharmac %>%
  filter(District %in% c("Capital & Coast", "Hutt Valley"))

unchanged_data <- pharmac %>%
  filter(!District %in% c("Capital & Coast", "Hutt Valley"))

merged_districts <- to_merge %>%
  group_by(YearDisp, TherapeuticGrp2) %>%
  summarise(
    NumDisps = sum(NumDisps, na.rm = TRUE),
    NumPpl = sum(NumPpl, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  mutate(District = "Capital, Coast and Hutt Valley") # Exact Census name

pharmac_clean <- bind_rows(unchanged_data, merged_districts)
```

```
integrated_data <- pharmac_clean %>%
  filter(District != "New Zealand") %>%
  left_join(
    census %>% filter(Area != "New Zealand"),
    by = c("District" = "Area")
  )

if ("Population" %in% colnames(integrated_data)) {
  integrated_data <- integrated_data %>%
    mutate(
      Dispensings_Per_1k = (NumDisps / Population) * 1000,
      Patients_Per_1k    = (NumPpl / Population) * 1000
    )
}
```

```
head(integrated_data)
```

##	TherapeuticGrp2					District	YearDisp
## 1	Agents Affecting the Renin-Angiotensin System					Auckland	2021
## 2	Agents Affecting the Renin-Angiotensin System					Bay of Plenty	2021
## 3	Agents Affecting the Renin-Angiotensin System					Canterbury	2021
## 4	Agents Affecting the Renin-Angiotensin System					Counties Manukau	2021
## 5	Agents Affecting the Renin-Angiotensin System					Hawke's Bay	2021
## 6	Agents Affecting the Renin-Angiotensin System					Lakes	2021
##	NumDisps	NumPpl	Population	Age_Under15	Age_15to29	Age_30to64	Age_Over65
## 1	208874	55118	480438	73830	109974	219471	61671
## 2	135111	34926	267177	51288	44232	114555	53757
## 3	266828	69100	598446	102996	116259	270321	99375
## 4	299183	78608	591720	128781	124209	263157	69045
## 5	95664	24700	177804	34620	30222	76212	33912
## 6	58327	15446	118608	23598	20466	50406	19881
##	Ethnicity_European		Ethnicity_Maori		Ethnicity_Pasifika		Ethnicity_Asian
## 1	239460		43455		61647		156603
## 2	197574		71199		9576		22002
## 3	468255		63204		22023		82557
## 4	216600		96411		157086		188715
## 5	128352		50037		10899		11343
## 6	75783		44649		6198		10068
##	Ethnicity_MELAAF		Ethnicity_Other		Maori_Descent		No_Maori_Descent
## 1	17229		3645		49857		406002
## 2	3456		2853		77559		178719
## 3	9657		7248		72648		498252
## 4	10494		5169		104874		467121
## 5	1398		2115		53463		115827
## 6	813		1269		47040		63900
##	Gender_Female		Gender_Other		Children_Average		Smoking_Regular
## 1	233430		2424		1.2		21606
## 2	134301		717		1.8		17013
## 3	295155		2691		1.5		32556
## 4	292968		1449		1.7		40131
## 5	89004		516		1.8		14544
## 6	57540		330		1.8		9096
##	Smoking_NonSmoker		Not_Disabled		Disabled		Income
## 1	297486		361407		22542		47700
## 2	134763		184431		16380		39100
## 3	332958		462123		35256		41800
## 4	329874		408966		31152		41300
## 5	84117		113940		11019		39300
## 6	54492		74865		7803		39300
##	Employment_PartTime		Employment_Unemployed		Employment_NotInLabourForce		
## 1	49482		12708		117561		
## 2	29922		6735		74250		
## 3	71928		11673		153825		
## 4	48576		17589		150531		
## 5	18579		3945		48039		
## 6	12429		3312		29592		
##	HoursWorked		Dispensings_Per_1k		Patients_Per_1k		
## 1	37.4		434.7575		114.7245		
## 2	36.6		505.6985		130.7223		
## 3	36.6		445.8681		115.4657		
## 4	38.2		505.6158		132.8466		

## 5	37.4	538.0306	138.9170
## 6	37.4	491.7628	130.2273

final attempt

```
top10_med_groups <- integrated_data %>%
  group_by(TherapeuticGrp2) %>%
  summarise(Grand_Total_Patients = sum(NumPpl, na.rm = TRUE), .groups = "drop") %>%
  slice_max(Grand_Total_Patients, n = 10, with_ties = FALSE) %>%
  pull(TherapeuticGrp2)

top10_integrated <- integrated_data %>%
  filter(TherapeuticGrp2 %in% top10_med_groups)
```

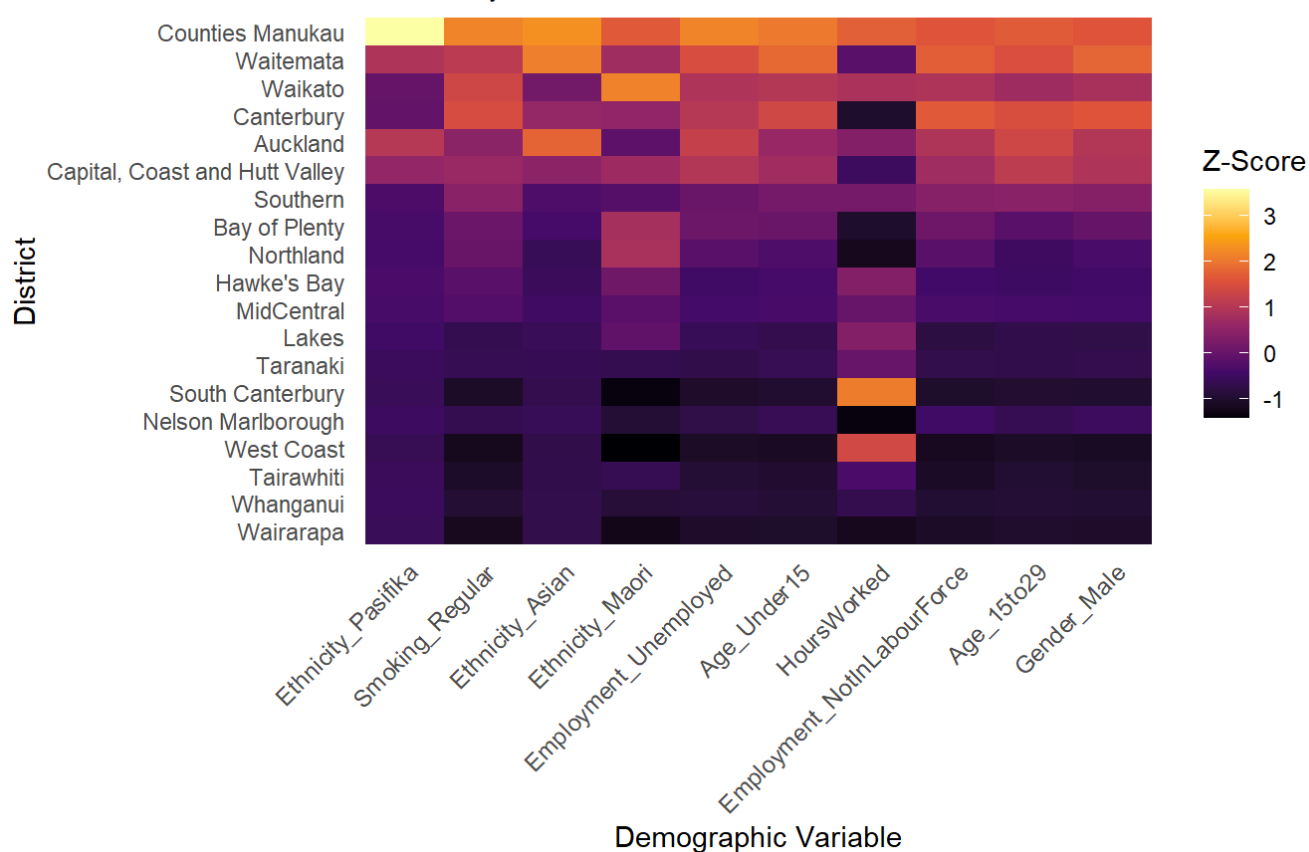
```
district_demo_profiles <- top10_integrated %>%
  select(District, Patients_Per_1k, where(is.numeric)) %>%
  select(-any_of(c("Total_Population_Column", "YearDisp", "NumDisps", "NumPpl", "Dispensing_Per_1k"))) %>%
  pivot_longer(cols = -c(District, Patients_Per_1k), names_to = "Demographic_Variable", values_to = "Demographic_Value") %>%
  group_by(District, Demographic_Variable) %>%
  summarise(Mean_Patient_Rate = mean(Patients_Per_1k, na.rm = TRUE), Mean_Demo_Val = mean(Demographic_Value, na.rm = TRUE), .groups = "drop")

top10_demo_correlations <- district_demo_profiles %>%
  group_by(Demographic_Variable) %>%
  summarise(Patient_Correlation = cor(Mean_Patient_Rate, Mean_Demo_Val, use = "complete.obs"), .groups = "drop") %>%
  slice_max(abs(Patient_Correlation), n = 10, with_ties = FALSE) %>%
  pull(Demographic_Variable)

district_demo_profiles %>%
  filter(Demographic_Variable %in% top10_demo_correlations) %>%
  group_by(Demographic_Variable) %>%
  mutate(Scaled_Value = scale(Mean_Demo_Val)[, 1]) %>%
  ungroup() %>%
  mutate(Demographic_Variable = factor(Demographic_Variable, levels = top10_demo_correlations), District = fct_reorder(District, Scaled_Value, .fun = sum)) %>%
  ggplot(aes(x = Demographic_Variable, y = District, fill = Scaled_Value)) +
  geom_tile(linewidth = 0.1) +
  scale_fill_viridis_c(option = "inferno", name = "Z-Score") +
  labs(title = "Top 10 Demographic Profiles in Districts", subtitle = "Ordered by correlation with Patient Rates", x = "Demographic Variable", y = "District") +
  theme_minimal() + theme(axis.text.x = element_text(angle = 45, hjust = 1, size = 9), panel.grid = element_blank())
```

Top 10 Demographic Profiles in Districts

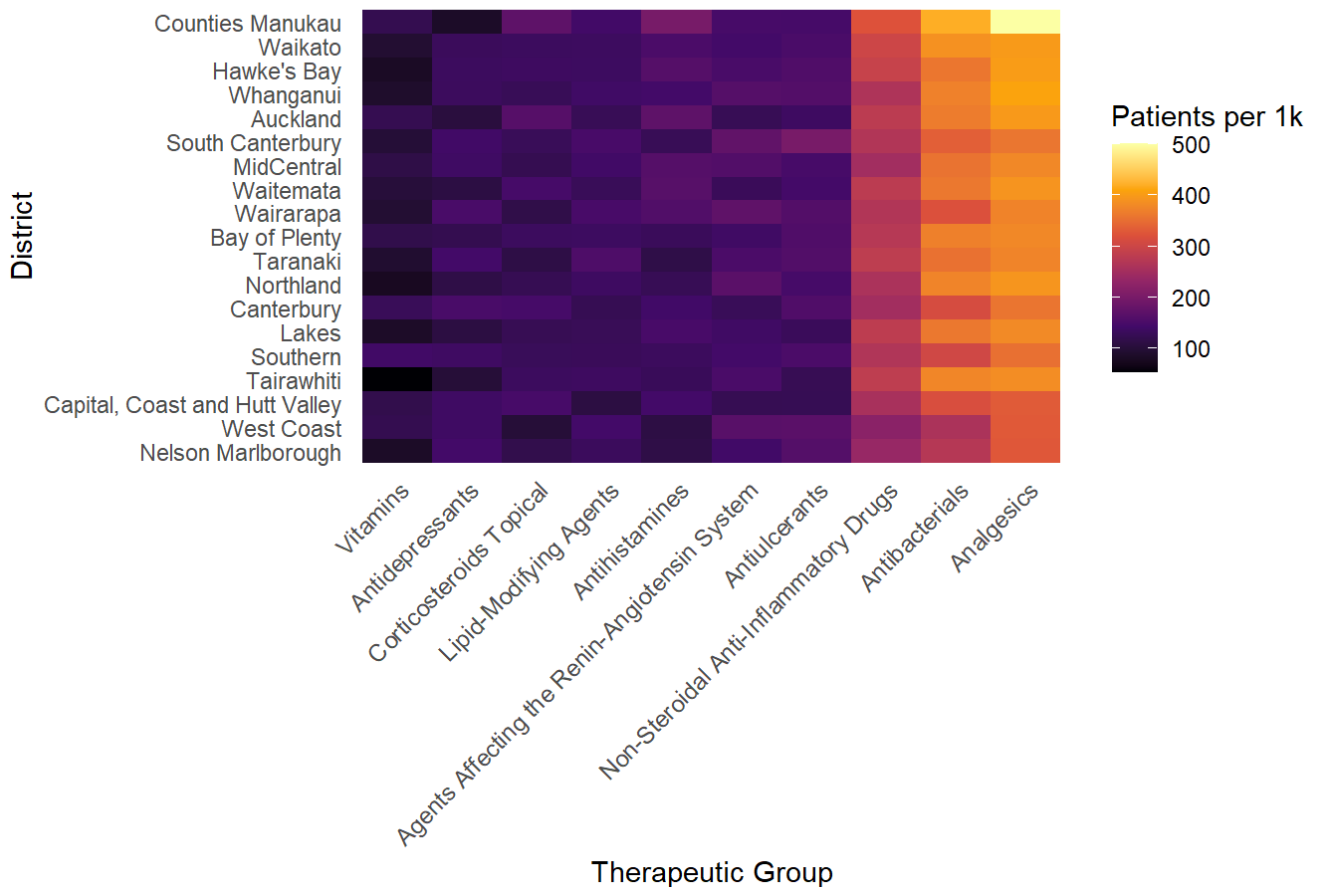
Ordered by correlation with Patient Rates



```
district_med_profiles <- top10_integrated %>%
  group_by(District, TherapeuticGrp2) %>%
  summarise(Mean_Patient_Rate = mean(Patients_Per_1k, na.rm = TRUE), .groups = "drop") %>%
  mutate(District = fct_reorder(District, Mean_Patient_Rate, .fun = sum), TherapeuticGrp2
= fct_reorder(TherapeuticGrp2, Mean_Patient_Rate, .fun = sum))

ggplot(district_med_profiles, aes(x = TherapeuticGrp2, y = District, fill = Mean_Patient_R
ate)) +
  geom_tile(linewidth = 0.1) +
  scale_fill_viridis_c(option = "inferno", name = "Patients per 1k") +
  labs(title = "Top 10 Medication Patient Rates by District", x = "Therapeutic Group", y =
"District") +
  theme_minimal() + theme(axis.text.x = element_text(angle = 45, hjust = 1, size = 9), pan
el.grid = element_blank())
```

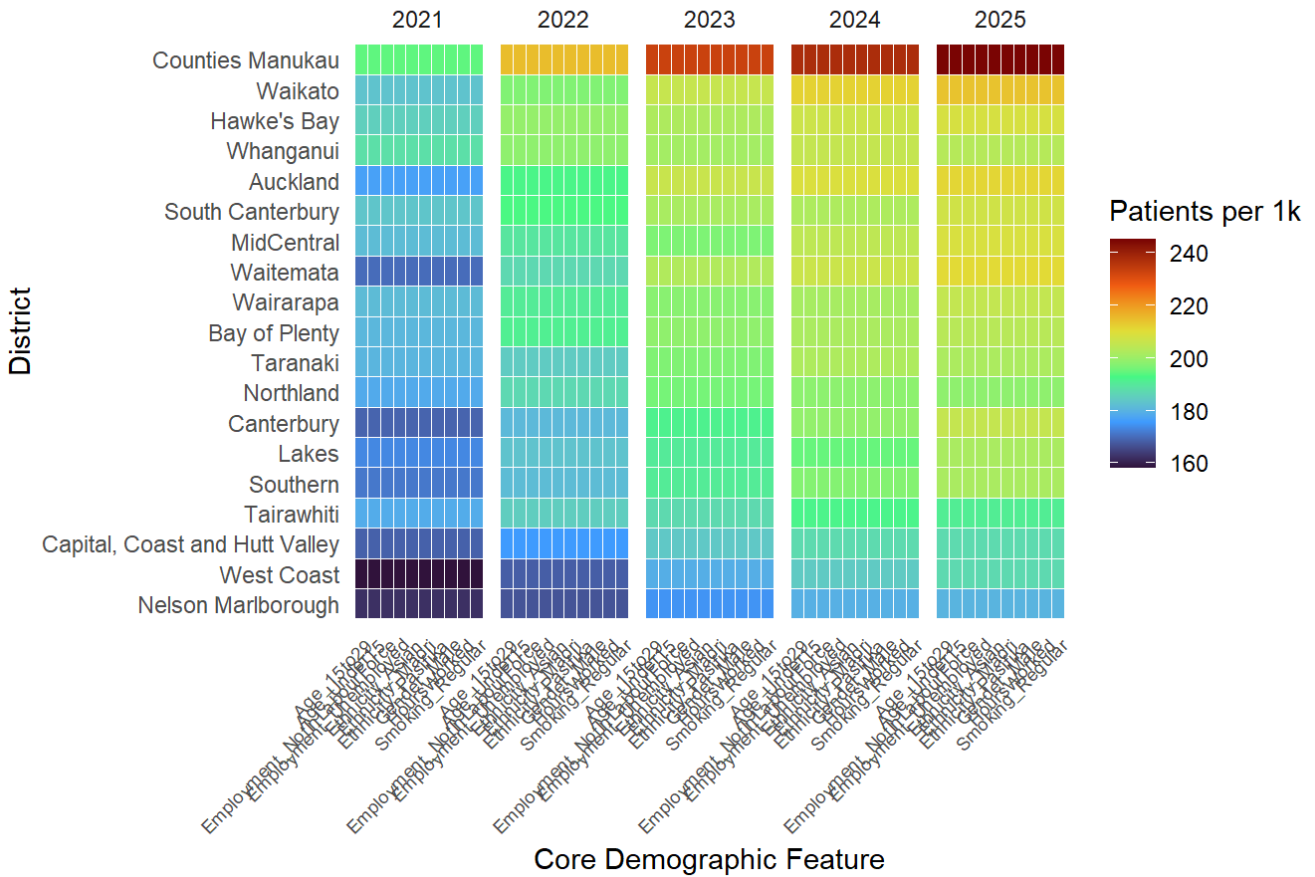
Top 10 Medication Patient Rates by District



```
annual_demo_shifts <- top10_integrated %>%
  select(District, YearDisp, Patients_Per_1k, where(is.numeric)) %>%
  select(-any_of(c("Total_Population_Column", "NumDisps", "NumPpl", "Dispensings_Per_1k"))) %>%
  pivot_longer(cols = -c(District, YearDisp, Patients_Per_1k), names_to = "Demographic_Variable", values_to = "Demographic_Value") %>%
  filter(Demographic_Variable %in% top10_demo_correlations) %>%
  group_by(District, YearDisp, Demographic_Variable) %>%
  summarise(Annual_Patient_Rate = mean(Patients_Per_1k, na.rm = TRUE), .groups = "drop") %>%
  mutate(District = fct_reorder(District, Annual_Patient_Rate, .fun = sum))

ggplot(annual_demo_shifts, aes(x = Demographic_Variable, y = District, fill = Annual_Patient_Rate)) +
  geom_tile(color = "white", linewidth = 0.1) +
  scale_fill_viridis_c(option = "turbo", name = "Patients per 1k") +
  facet_wrap(~ YearDisp, ncol = 5) +
  labs(title = "Demographic Patient Rate Shifts Across Years", x = "Core Demographic Feature", y = "District") +
  theme_minimal() + theme(axis.text.x = element_text(angle = 45, hjust = 1, size = 7), panel.grid = element_blank())
```

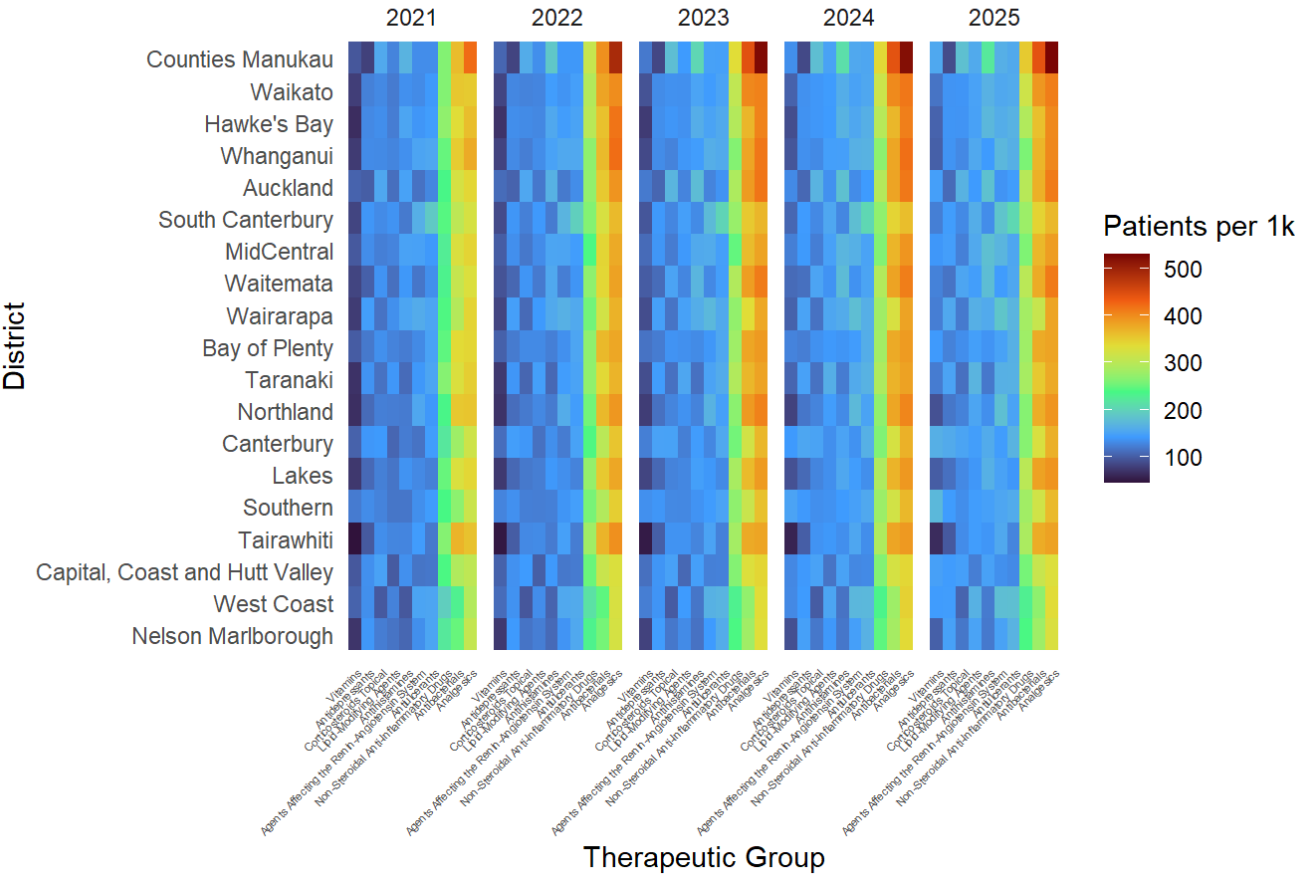
Demographic Patient Rate Shifts Across Years



```
# Aggregate granular medication rates per year
annual_med_shifts <- top10_integrated %>%
  group_by(District, YearDisp, TherapeuticGrp2) %>%
  summarise(Annual_Patient_Rate = mean(Patients_Per_1k, na.rm = TRUE), .groups = "drop")
%>%
  mutate(District = fct_reorder(District, Annual_Patient_Rate, .fun = sum), TherapeuticGrp
2 = fct_reorder(TherapeuticGrp2, Annual_Patient_Rate, .fun = sum))

ggplot(annual_med_shifts, aes(x = TherapeuticGrp2, y = District, fill = Annual_Patient_Rat
e)) +
  geom_tile(linewidth = 0.1) +
  scale_fill_viridis_c(option = "turbo", name = "Patients per 1k") +
  facet_wrap(~ YearDisp, ncol = 5) +
  labs(title = "Medication Patient Rate Shifts Across Years", x = "Therapeutic Group", y =
"District") +
  theme_minimal() + theme(axis.text.x = element_text(angle = 45, hjust = 1, size = 4), pan
el.grid = element_blank())
```

Medication Patient Rate Shifts Across Years



I also want medication and demographic